Chapter 2: Prompt Engineering Basics

Class 12: Practical / Activities (10 Minutes)

Topic: Prompt Engineering Practice, Revision & Practical Assessment

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Activity 1: Prompt Engineering Skills Revision (4 Minutes)

Objective

To help students revise all the prompt engineering concepts learned in this chapter by writing, testing, and improving AI prompts for different educational tasks.

Materials Required

- Computer/Laptop or Tablet
- Internet Connection
- AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, etc.)
- Student Revision Worksheet
- Notebook & Pen
- Teacher Observation Sheet

Instructions

1. Open the AI tool.
2. Select one topic from the revision list.
3. Complete the following prompt engineering tasks:
   - Write a **text prompt**.
   - Write an **image prompt**.
   - Write a **presentation prompt**.
   - Improve one weak prompt.
4. Generate AI responses for each prompt.
5. Review the responses and identify which prompt produced the best result.

Suggested Revision Topics

- Artificial Intelligence
- Water Conservation
- Climate Change
- Renewable Energy
- Cyber Safety
- Digital Citizenship
- Healthy Lifestyle
- Space Exploration
- Environmental Protection
- Plastic-Free Earth

Student Revision Worksheet

Task	Completed (✓)
Text Prompt Written	☐
Image Prompt Written	☐
Presentation Prompt Written	☐
Weak Prompt Improved	☐
AI Responses Reviewed	☐
Best Prompt Identified	☐

Student Activity Checklist

Students should:

✔ Write different types of prompts.

✔ Use clear instructions.

✔ Include audience and output format.

✔ Improve weak prompts.

✔ Review AI-generated responses.

✔ Complete the worksheet.

Teacher Observation Checklist

- [x] Revises all prompt engineering skills.
- [x] Writes effective prompts.
- [x] Uses AI responsibly.
- [x] Demonstrates independent learning.

Activity 2: Prompt Engineering Practical Challenge (3 Minutes)

Objective

To assess students' ability to apply prompt engineering skills by creating high-quality prompts for different AI tasks within a limited time.

Materials Required

- AI Tool
- Practical Challenge Worksheet
- Teacher Evaluation Sheet

Instructions

The teacher assigns one topic.

Students must complete the following tasks within **3 minutes**:

- Write one **text prompt**.
- Write one **image prompt**.
- Write one **presentation prompt**.
- Improve one weak prompt.
- Generate AI responses and compare the results.

Challenge Worksheet

Task	Student Response
Topic	______________________________
Text Prompt	______________________________
Image Prompt	______________________________
Presentation Prompt	______________________________
Improved Prompt	______________________________
Best AI Response	______________________________

Discussion Questions

1. Which type of prompt was easiest to write?
2. Which prompt produced the best AI response?
3. Why did the improved prompt perform better?
4. How can prompt engineering improve productivity?
5. Why is prompt refinement important?

Teacher Observation Checklist

- [x] Completes the challenge within the given time.
- [x] Demonstrates prompt engineering skills.
- [x] Uses AI effectively.
- [x] Participates confidently.

Activity 3: Practical Assessment & Self-Reflection (3 Minutes)

Objective

To evaluate students' understanding of prompt engineering concepts and encourage self-reflection on effective and responsible AI usage.

Materials Required

- Practical Assessment Sheet
- Self-Assessment Checklist
- Teacher Observation Sheet

Instructions

Students review their completed work and answer the following reflection questions.

Self-Assessment Checklist

Skill	Yes	Needs Practice
I can write a clear text prompt.	☐	☐
I can create an AI image prompt.	☐	☐
I can write a presentation prompt.	☐	☐
I can improve a weak prompt.	☐	☐
I can compare AI responses.	☐	☐
I can refine prompts for better results.	☐	☐
I can use AI responsibly and ethically.	☐	☐

Reflection Questions

1. Which prompt engineering skill did you enjoy the most?
2. Which type of prompt do you need to improve?
3. How will prompt engineering help you in your future studies?
4. What is one responsible AI practice you will always follow?

Teacher Observation Checklist

☑ Demonstrates understanding of all chapter concepts.

☑ Reflects honestly on learning progress.

☑ Uses AI responsibly.

☑ Completes the assessment independently.

Mini Practical Assessment Rubric

Assessment Area	Marks
Text Prompt Writing	20
Image Prompt Writing	20
Presentation Prompt Writing	20
Prompt Refinement & AI Response Evaluation	20
Overall Participation & Practical Skills	20
Total	**100 Marks**

Teacher Observation Checklist

☑ Demonstrates all prompt engineering skills confidently.

☑ Writes clear, structured, and detailed prompts.

☑ Improves prompts after evaluating AI responses.

☑ Uses AI tools responsibly and ethically.

☑ Participates actively throughout the practical assessment.

Practical Time Distribution

Activity	Duration
Prompt Engineering Skills Revision	**4 Minutes**
Prompt Engineering Practical Challenge	**3 Minutes**
Practical Assessment & Self-Reflection	**3 Minutes**
Total Practical Time	**10 Minutes**

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Apply prompt engineering techniques to create effective **text, image, and presentation prompts**.
- Improve weak prompts by adding **clarity, context, audience, output format, and specific instructions**.
- Compare and evaluate AI-generated responses to identify the most accurate and useful outputs.
- Demonstrate confidence in refining prompts to achieve better AI-generated results.
- Use AI responsibly by following ethical practices, verifying information, and respecting academic integrity.
- Integrate prompt engineering skills into school projects, presentations, research, and creative digital content development.

Chapter 2: Prompt Engineering Basics

Chapter-End Activity

Activity 1: Individual – Prompt Engineering Quiz & Mini Prompt Assessment

Objective

To assess each student's understanding of prompt engineering concepts, prompt writing techniques, prompt refinement, AI response evaluation, and responsible AI usage through a written quiz and practical prompt-writing task.

Materials Required

- Computer/Laptop or Tablet
- Internet Connection
- AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, etc.)
- Prompt Engineering Quiz Worksheet
- Practical Assessment Sheet
- Teacher Evaluation Sheet

Instructions

1. Students complete the **Prompt Engineering Quiz** individually.
2. After the quiz, each student completes one practical prompt-writing activity using an AI tool.
3. Students compare and improve their AI-generated responses.
4. The teacher evaluates both theoretical knowledge and practical prompt engineering skills.

Part A – Prompt Engineering Quiz

Multiple Choice Questions

1. What is a prompt?

a) A computer virus

b) An instruction or question given to an AI system ✅

c) A programming language

d) A computer game

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2. Which prompt is more effective?

a) Tell me about trees.

b) Explain the importance of trees for Class IX students using five bullet points and simple English. ✅

c) Trees.

d) Write something.

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3. Why should prompts include context?

a) To confuse AI

b) To make the response more accurate and relevant ✅

c) To increase typing

d) It is not necessary

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4. Which element is important in a good prompt?

a) Audience

b) Output Format

c) Clear Instructions

d) All of the Above ✅

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5. What should students do after receiving an AI response?

a) Copy it directly

b) Ignore it

c) Review and improve it if needed ✅

d) Delete it

Part B – Short Answer Questions

1. What is Prompt Engineering?
2. Write any four characteristics of a good prompt.
3. Why is prompt refinement important?
4. What is an image prompt?
5. Mention four responsible AI practices.

Mini Prompt Assessment

Each student completes **one** of the following tasks.

Option A – Text Prompt

Write a prompt that generates **five key points** about Climate Change.

Option B – Image Prompt

Write an image prompt to create a **World Environment Day Poster**.

Option C – Presentation Prompt

Write a prompt that creates a **5-slide presentation outline** on Artificial Intelligence.

Option D – Prompt Improvement

Improve the following prompt:

Tell me about computers.

Teacher Observation Checklist

- [x] Writes clear AI prompts.
- [x] Uses proper prompt structure.
- [x] Improves AI responses.
- [x] Demonstrates responsible AI usage.
- [x] Completes the assessment independently.

Activity 2: Team – Prompt Writing Challenge & AI Creativity Workshop

Objective

To encourage teamwork, creativity, and effective communication by designing high-quality prompts for different AI tasks.

Materials Required

- Computer/Laptop or Tablet
- Internet Connection
- AI Tool
- Prompt Challenge Worksheet
- Chart Paper or Canva
- Teacher Evaluation Sheet

Instructions

1. Divide students into **groups of 3–5 members**.
2. Each team selects one topic.
3. The team creates:
    - One Text Prompt
    - One Image Prompt
    - One Presentation Prompt
4. Generate AI responses.
5. Improve the prompts if necessary.
6. Present the prompts and AI outputs to the class.

Suggested Topics

- Artificial Intelligence
- Digital Safety
- Climate Change
- Water Conservation
- Renewable Energy
- Healthy Lifestyle
- Cyber Safety
- Space Exploration
- Plastic-Free Earth
- Environmental Protection

Workshop Tasks

Each team should:

- Write one effective **Text Prompt**.
- Write one **Image Prompt**.
- Write one **Presentation Prompt**.
- Improve one weak prompt.
- Compare AI-generated responses.
- Present the final prompts and explain why they are effective.

Group Discussion Questions

1. Which prompt produced the best AI response?
2. How did prompt refinement improve the output?
3. Which prompt was most creative?
4. Why is prompt engineering important in education?
5. How can AI help students learn more effectively?

Teacher Observation Checklist

✅ Collaborates effectively.

✅ Writes creative prompts.

✅ Improves AI responses.

✅ Explains prompt design confidently.

✅ Uses AI responsibly.

Activity 3: Recognition – Best Prompt Engineer Award

Objective

To recognize students who demonstrate outstanding prompt-writing skills, creativity, and responsible AI usage.

Award Criteria

Students are evaluated based on:

- Prompt Clarity
- Prompt Structure
- Creativity
- Context
- Output Format
- AI Response Quality
- Prompt Refinement
- Responsible AI Usage
- Presentation Skills
- Overall Performance

Recognition

🏆 Best Prompt Engineer Award

Awarded to the student who writes the most effective, creative, and well-structured AI prompts while demonstrating excellent prompt engineering skills.

Activity 4: Recognition – Best Prompt Design Team Award

Objective

To recognize the team that demonstrates outstanding collaboration, creativity, and prompt engineering skills.

Award Criteria

Teams are evaluated on:

- Teamwork
- Prompt Quality
- Creativity
- AI Response Quality
- Prompt Improvement
- Communication
- Presentation
- Equal Participation
- Time Management
- Overall Performance

Recognition

🏆 Best Prompt Design Team Award

Awarded to the team that demonstrates excellent collaboration, creativity, prompt refinement, and high-quality AI-generated outputs.

Activity 5: Practical Evaluation

Objective

To evaluate students' practical understanding of prompt engineering concepts and their ability to create effective prompts for different AI applications.

Practical Tasks

Students should demonstrate the ability to:

- Write clear text prompts.
- Create image prompts.
- Write presentation prompts.
- Improve weak prompts.
- Compare AI responses.
- Refine prompts.
- Evaluate AI-generated outputs.

- Use AI responsibly.
- Present prompt-writing strategies.
- Apply prompt engineering in real-life learning tasks.

Practical Evaluation Rubric

Assessment Area	Marks
Text Prompt Writing	10
Image Prompt Writing	10
Presentation Prompt Writing	10
Prompt Refinement	10
AI Response Evaluation	10
Creativity & Innovation	10
Responsible AI Usage	10
Communication & Presentation	10
Team Collaboration	10
Overall Practical Performance	10
Total	**100 Marks**

Teacher Observation Checklist

☑ Writes clear, detailed, and effective prompts.

☑ Improves prompts using refinement techniques.

☑ Evaluates AI-generated responses critically.

☑ Demonstrates creativity and responsible AI practices.

☑ Communicates ideas confidently and works effectively in a team.

Chapter Learning Outcomes

After completing the **Chapter-End Activities** for **Chapter 2: Prompt Engineering Basics**, students will be able to:

- Write effective **text, image, and presentation prompts** for a variety of educational and creative tasks.
- Improve weak prompts by adding **clarity, context, audience, output format, and specific instructions**.
- Compare and evaluate AI-generated responses to identify the most accurate and useful results.
- Apply prompt engineering techniques to enhance learning, research, presentations, and digital content creation.
- Demonstrate ethical, responsible, and productive use of AI tools while protecting privacy and maintaining academic integrity.
- Collaborate effectively in team-based prompt engineering activities and confidently present AI-assisted solutions.